Grid Computing: Application to Science

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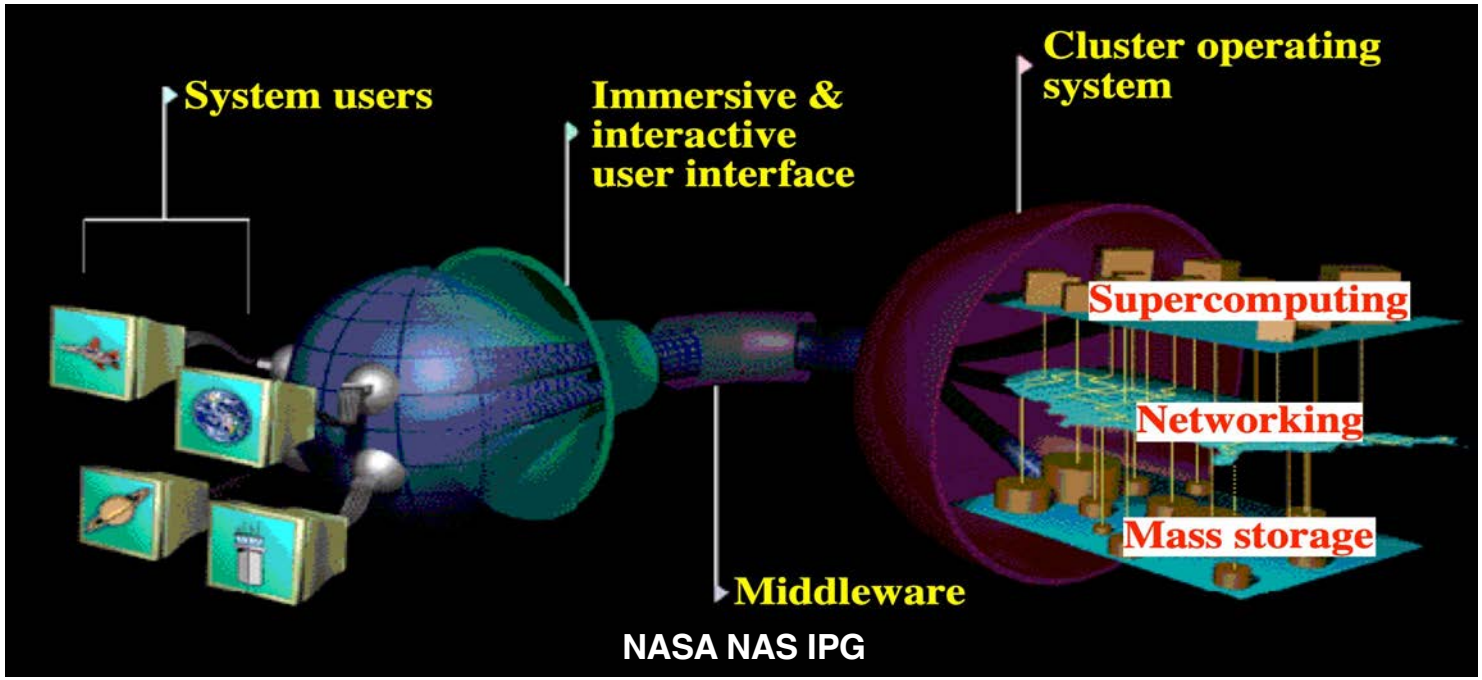
Email: anakano@usc.edu

Grid = gateway to post-Exascale (fault resilience, latency hiding) & cloud computing



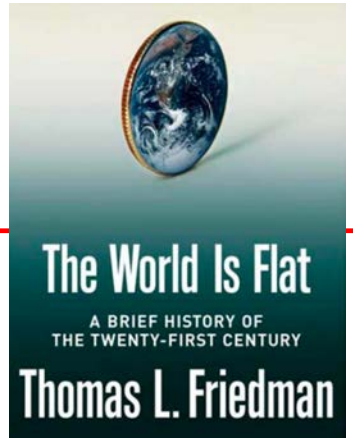
Grid Computing

- **World Wide Web:** Universal interface to digital library on the Internet
- **Information Grid:** Pervasive (from any place in the world at any time) access to everything (computing, mass storage, experimental equipment, distributed sensors, *etc*., on high-speed networks)



Scientific Grid Applications

1. **Distributed supercomputing (metacomputing):** Uses geographically distributed multiple supercomputers to tackle problems that cannot be solved on a single platform
2. **Data-intensive science:** Synthesizes new knowledge from massive data maintained in geographically distributed repositories, digital libraries & databases (Google science)
3. **Remote experimentation:** Teleoperation & teleobservation of experiments
4. **Collaborative computing:** Enable human-to-human interactions in a virtual shared space (*cf.* metaverse)



- **Virtual community science** democratization of science: "Do I really need all that infrastructure to do science?"

Application-Level Grid Tools

Grid programming models

- **Message passing: MPICH-G2**
- **Remote procedure call: Ninf-G**

Grid application types

- **Metacomputing**
- **Parameter-sweep (high throughput) applications**
- **Workflow applications**
- **Portals: Thin-client, graphical user interfaces to the Grid**

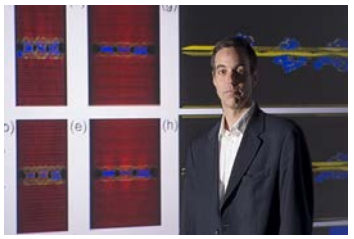


Ewa Deelman

cf. https://pegasus.isi.edu

cf. https://nanohub.org

Alejandro Strachan

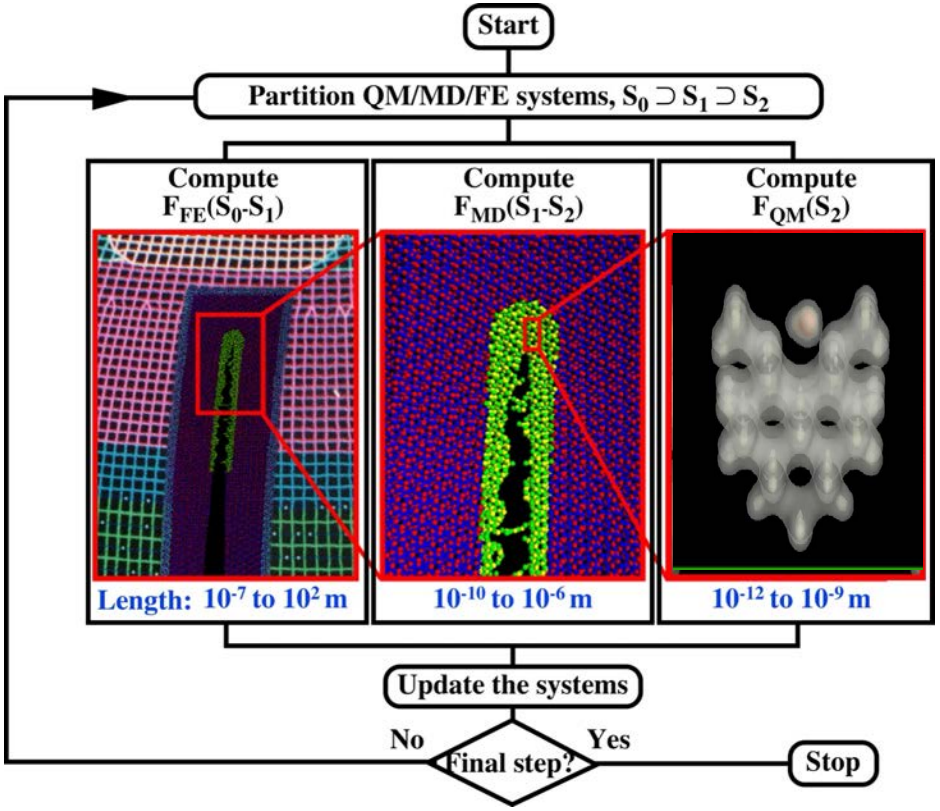


Outline

1. Grid programming
   - > Metacomputing—multiscale MD/quantum-mechanical (QM) simulations: Grid-enabled MPI (MPI–G2)
   - > Task farm: Grid remote procedure call (Ninf–G)
   - > Sustainable & adaptive Grid supercomputing
2. Grid software
   - > Globus toolkit
   - > Open Grid Services Architecture (OGSA)

Multiscale FE/MD/QM Simulation

- Embed high-accuracy computations only when & where needed
- Train coarse simulations by fine simulations



Multiscale simulation to seamlessly couple:

- Finite-element (FE) calculation based on continuum elasticity
- Atomistic molecular-dynamics (MD) simulation
- Quantum-mechanical (QM) calculation based on the density functional theory (DFT)

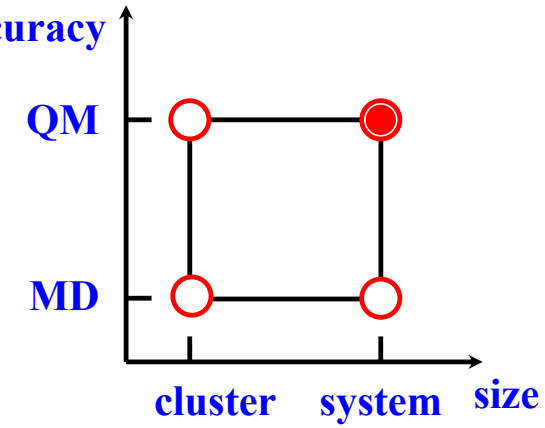
Grid-Enabled MD/QM Algorithm

Additive hybridization Morokuma *et al.*, '96

- **Extrapolation in meta-model space (accuracy *vs.* size)**

$$E \cong E_{\mathrm{MD}}^{\mathrm{system}} + E_{\mathrm{QM}}^{\mathrm{cluster}} - E_{\mathrm{MD}}^{\mathrm{cluster}}$$



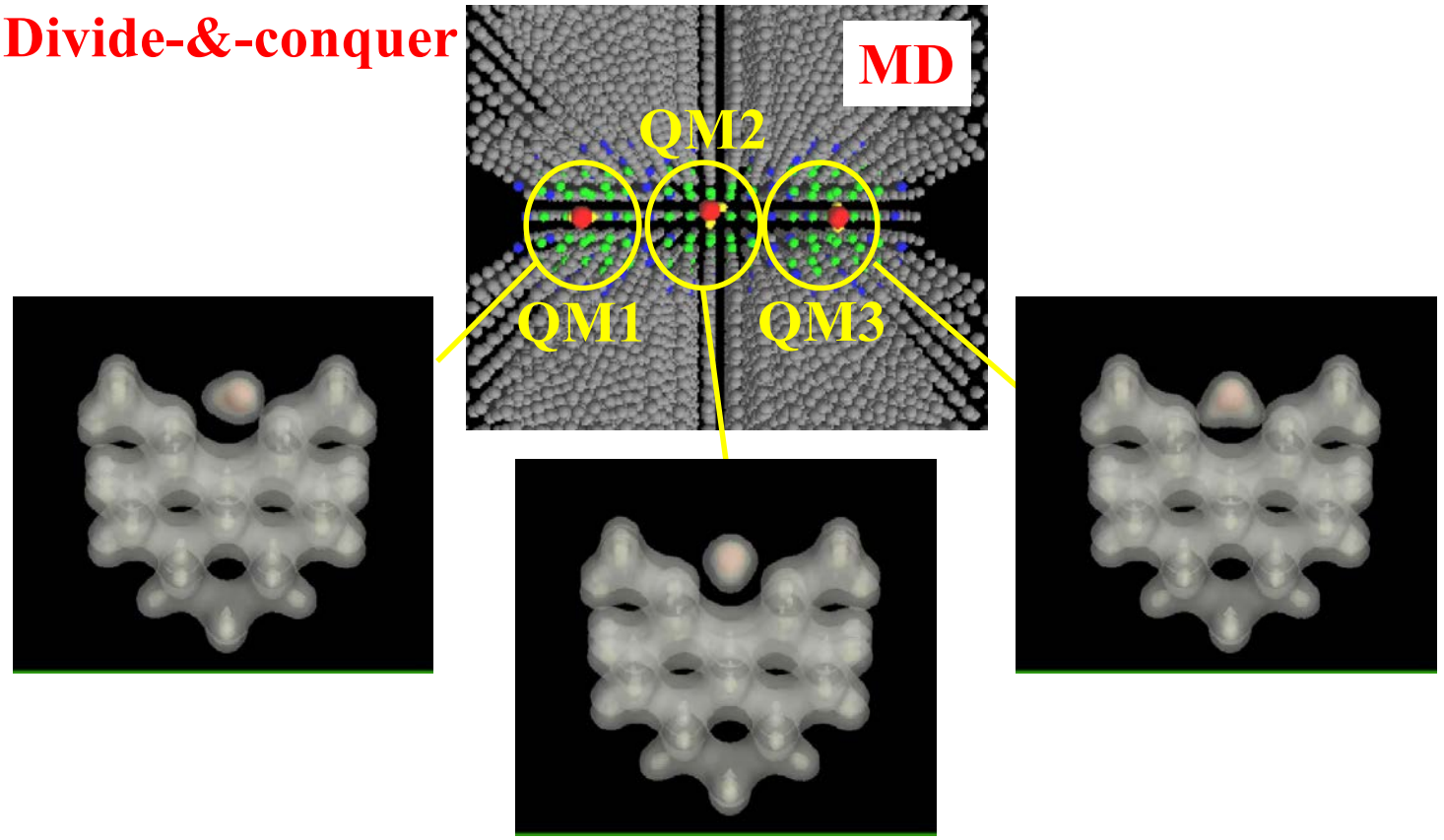


- **Modular**
  - → **Reuse of existing MD & QM codes**
  - → **Minimal inter-model dependence/communication**

Grid Enabling: Multiple QM Clustering

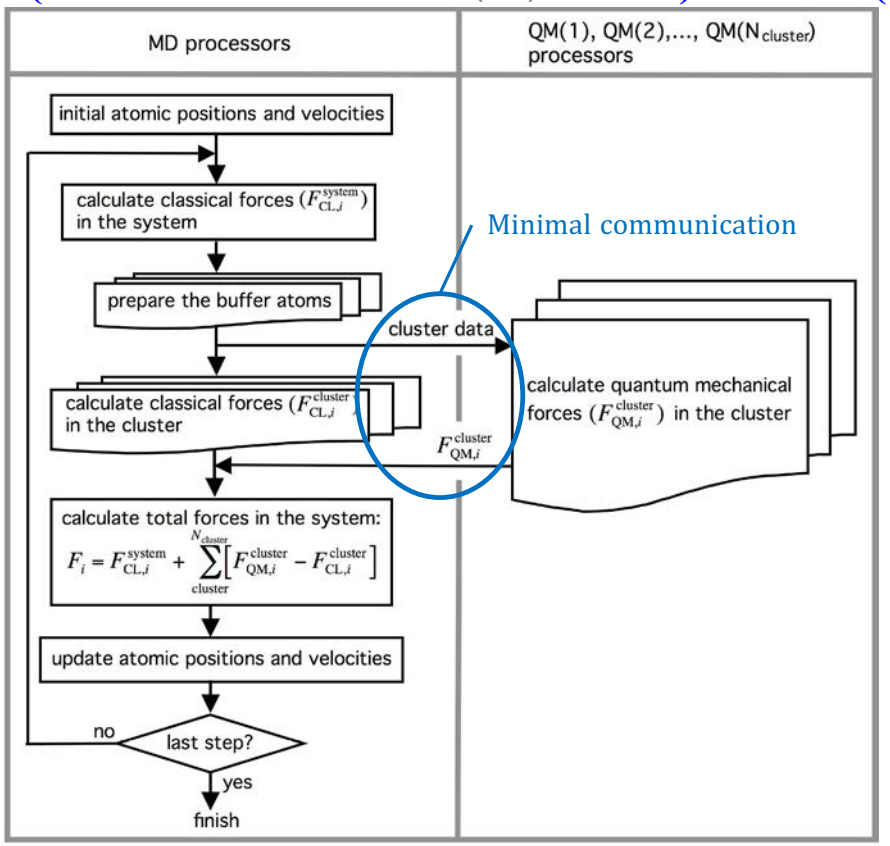
$$E = E_{\mathrm{MD}}^{\mathrm{system}} + \sum_{\mathrm{cluster}} [E_{\mathrm{QM}}^{\mathrm{cluster}}(\{\mathbf{r}_{\mathrm{QM}}\},\{\mathbf{r}_{\mathrm{HS}}\}) - E_{\mathrm{MD}}^{\mathrm{cluster}}(\{\mathbf{r}_{\mathrm{QM}}\},\{\mathbf{r}_{\mathrm{HS}}\})]$$

Divide-&-conquer



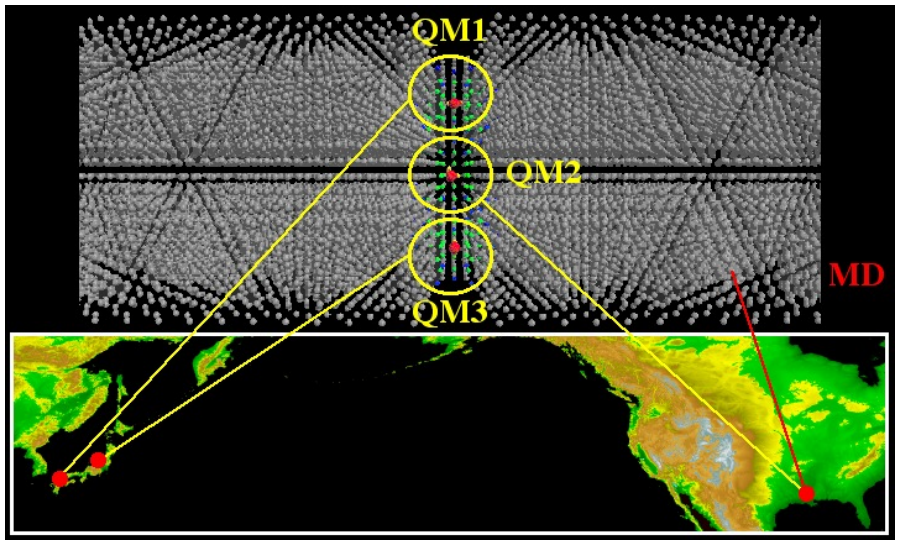
Grid Implementation

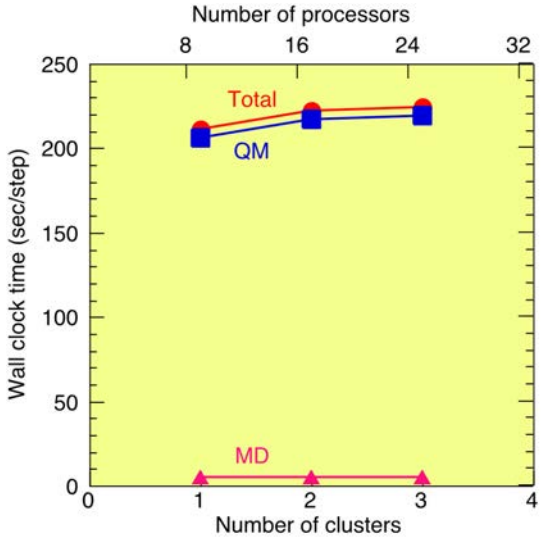
- **Task decomposition (MPI Communicator) + spatial decomposition**
- **Computation/communication overlapping to hide latency** MPI_Irecv()
- **MPICH-G2 (**doi: 10.1016/S0743-7315(03)00002-9**)/Globus (**www.globus.org**)**



Global Collaborative Simulation

Hybrid MD/QM simulation on a Grid of distributed PC clusters in the US & Japan





Japan: Yamaguchi — 65 P4 2.0GHz
Hiroshima, Okayama, Niigata — 3×24 P4 1.8GHz
US: Louisiana — 17 Athlon XP 1900+

MD — 91,256 atoms
QM (DFT) — 76*n* atoms on *n* clusters

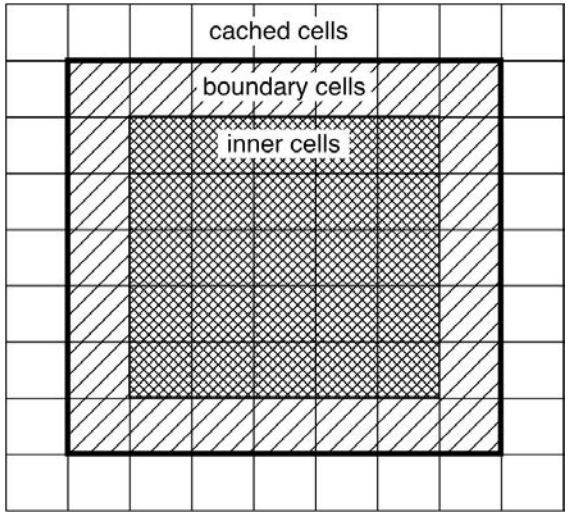
- **Scaled speedup, $P = 1$ (for MD) + $8n$ (for QM)**
- **Weak-scaling efficiency = 0.94 on 25 processors over 3 PC clusters**

H. Kikuchi *et al.*, *IEEE/ACM SC02*

Grid-Enabled MD Algorithm

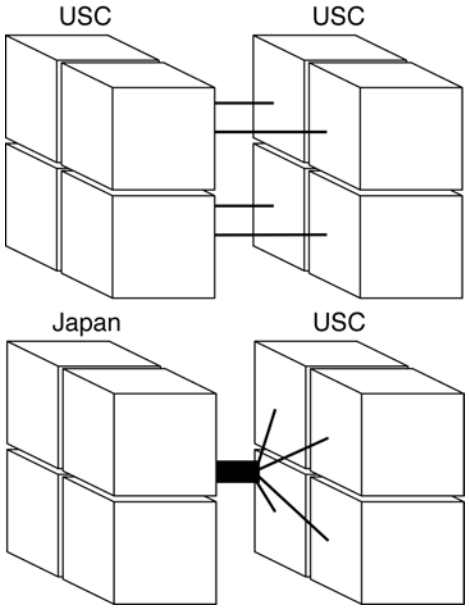
Grid MD algorithm:

1. asynchronous receive of cells to be cached MPI_Irecv()
2. send atomic coordinates in the boundary cells
3. compute forces for atoms in the inner cells
4. wait for the completion of the asynchronous receive MPI_Wait()
5. compute forces for atoms in the boundary cells



Renormalized Messages:

Latency can be reduced by composing a large cross-site message instead of sending all processor-to-processor messages



Fast TCP

BBC NEWS WORLD EDITION

Last Updated: Thursday, 5 June, 2003, 10:48 GMT 11:48 UK

E-mail this to a friend Printable version

Promise of ultra-fast downloads

Soon you could be downloading an entire movie off the net far faster than you do now.



Fast net tech could soon take off

US researchers are working on ways to improve the way that net protocols decide how quickly data travels around the net.

Early tests of the new system show that it can triple data transmission speeds.

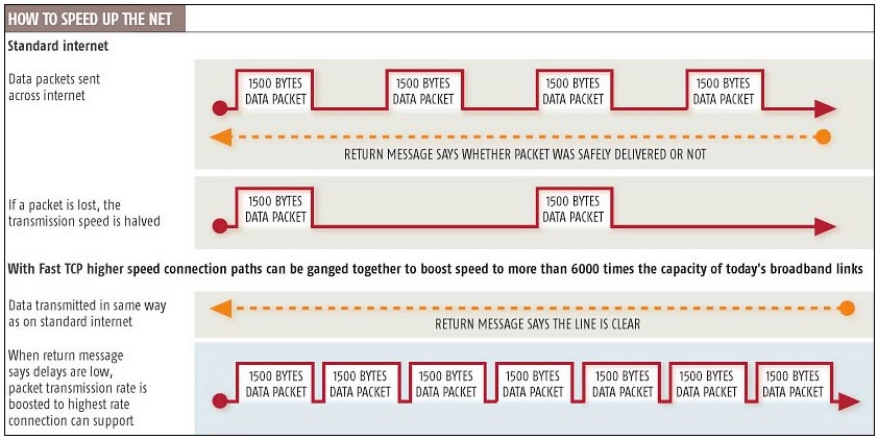
By linking lots of the faster systems together the researchers have produced data transfer speeds many times higher than is possible today.

Packet tracking promises ultrafast internet

10:54 05 June 03

Exclusive from New Scientist Print Edition. Subscribe and get 4 free issues.

Imagine an internet connection so fast it will let you download a whole movie in just five seconds, or access TV-quality video servers in real time. That is the promise from a team at the California Institute of Technology in Pasadena, who have developed a system called Fast TCP.



Fast TCP: Achieved 8.6 Gb/s between Sunnyvale, CA & CERN, Switzerland

Steven Low (Caltech)

http://netlab.caltech.edu

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   - > Open Grid Services Architecture (OGSA)

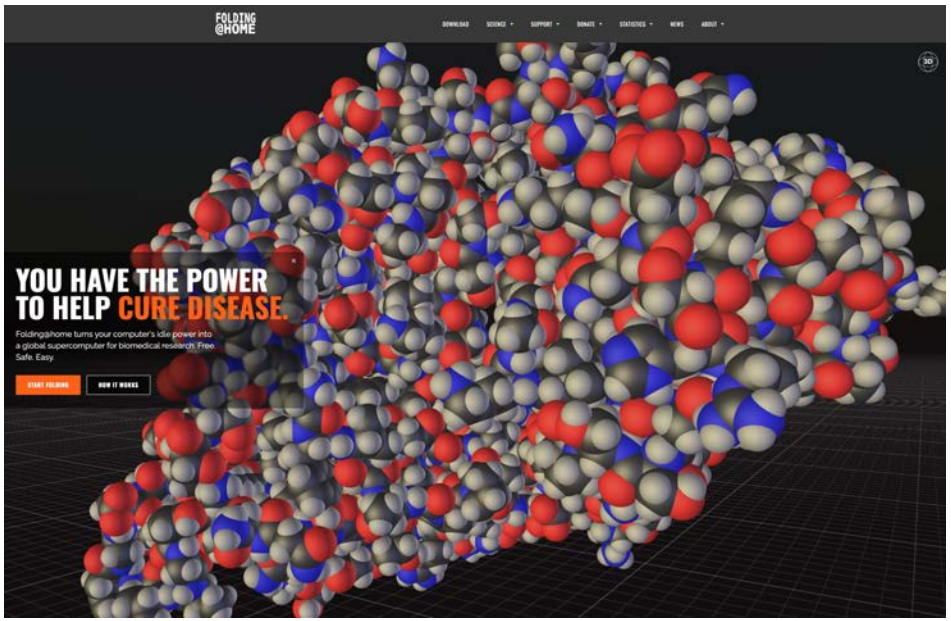
Task Farm Applications



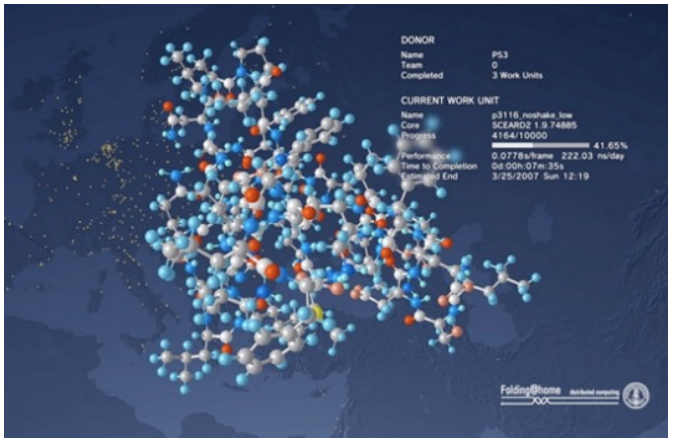
cf. Protein folding in 2024 Nobel Chemistry Prize

Number CPUs	Number Active CPUs	Number Users	Number Teams	Last Update
423995	90438	210257	25971	2003-05-05 20:04:04

OS type	Active	Total
Windows	86473	369859
Mac OS X	2653	24129
Linux	1294	29931
Other	0	13
Total	90420	423932



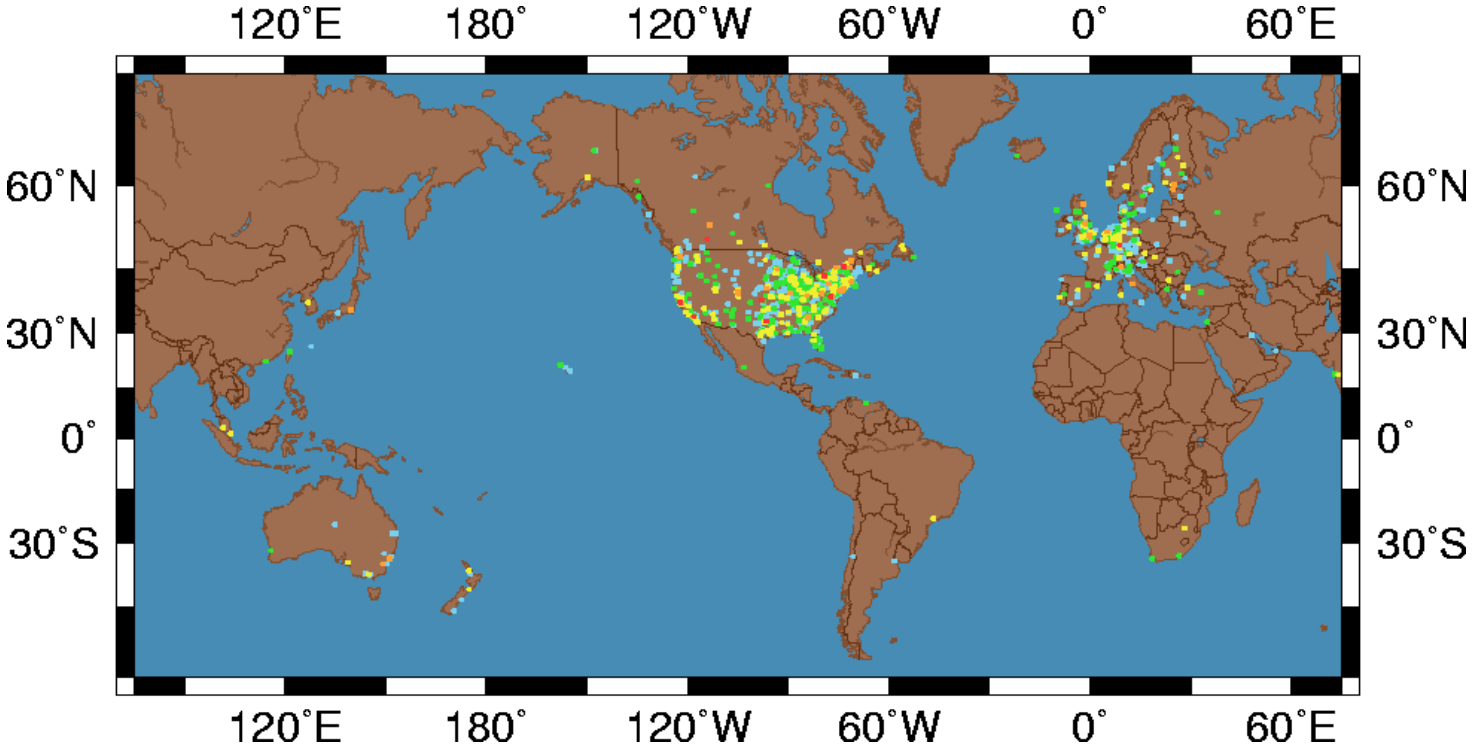
Screen saver
(*cf.* OpenGL idle event handler)



https://foldingathome.org

World Wide Distributed Computing

Folding@home



https://foldingathome.org

Enabling Science by Online Game

nature Vol 466|5 August 2010|doi:10.1038/nature09304

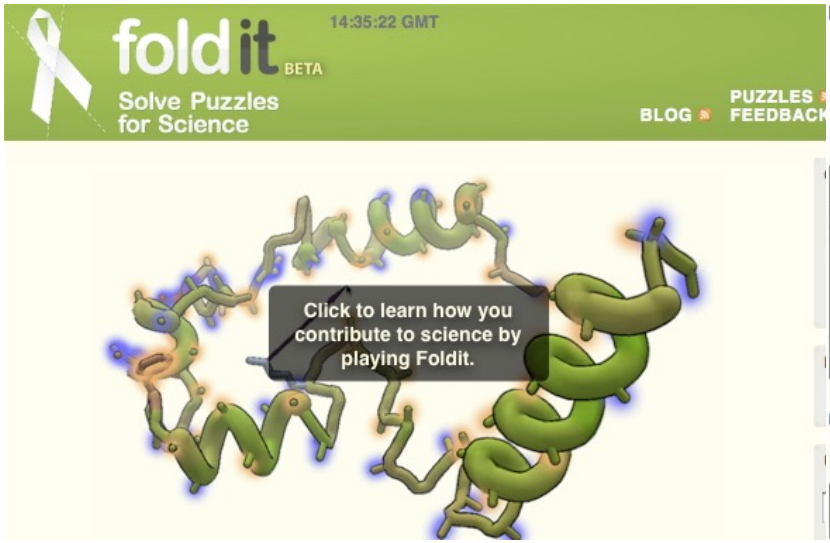
LETTERS

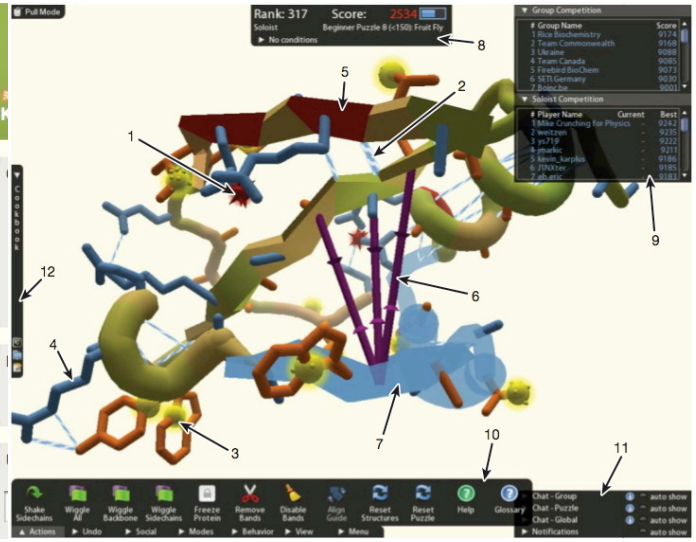
From passive to active users

Predicting protein structures with a multiplayer online game

2024 Nobel Chemistry Prize

Seth Cooper[1], Firas Khatib[2], Adrien Treuille[1,3], Janos Barbero[1], Jeehyung Lee[3], Michael Beenen[1], Andrew Leaver-Fay[2]†, David Baker[2,4], Zoran Popović[1] & Foldit players





Quake-Catcher Network

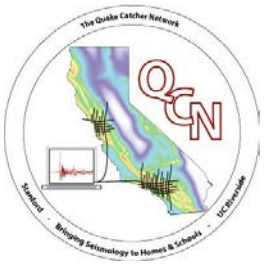
- **Network of accelerometer-equipped laptops/desktops for early earthquake warning & research**
- **Clustering accelerometer time series data to detect earthquakes**

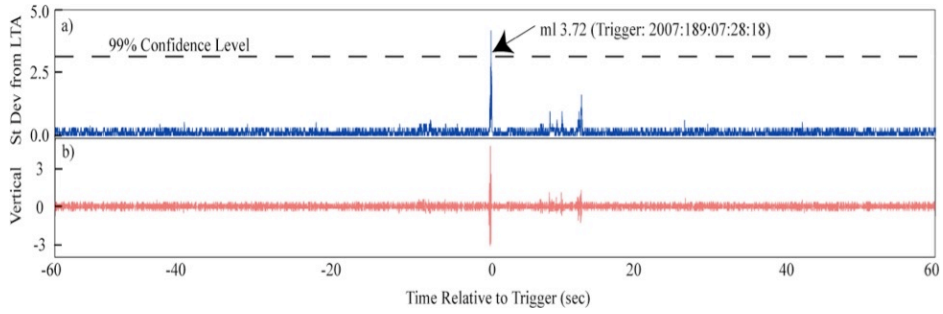




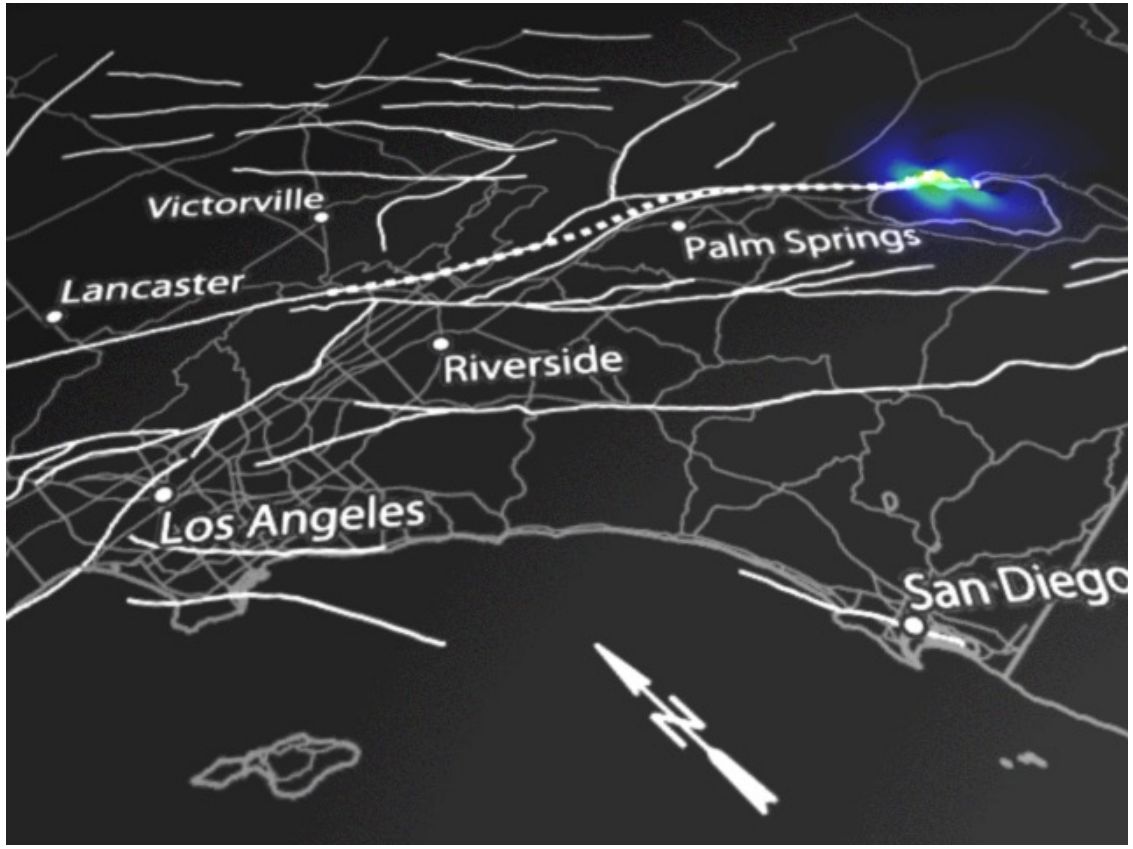


https://en.wikipedia.org/wiki/Quake-Catcher_Network





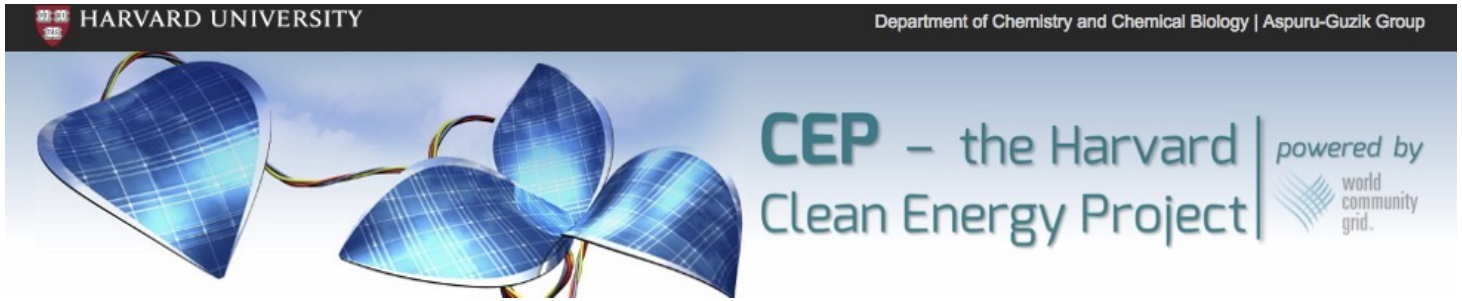
Virtual Earthquake: Atomic to Tectonic



Southern California Earthquake Center (SCEC)

http://www.scec.org

Harvard Clean Energy Project



http://cleanenergy.molecularspace.org

world community grid.
technology solving problems

What's New

World Community Grid: now available on Android!

Your smartphone or tablet can now help search for the next HIV treatment. Join today or add your Android device to your existing account.

Learn more

Who We Are

World Community Grid brings together people from across the globe to benefit humanity by creating the world's largest non-profit computing grid. We do this by pooling surplus processing power from volunteers' devices. We believe that innovation combined with visionary scientific research and large-scale volunteerism can help make the planet smarter. Our success depends on like-minded individuals - like you.

How You Can Help

Download and install secure, free software that captures the spare processing power of your computer, smartphone or tablet, and harnesses it for scientific research.

Join Today!

What's New

The Clean Energy Project data published!

Harvard researchers published their database with the electronic properties of 2.3 million organic compounds. What might this mean for the future of solar cells?

http://www.worldcommunitygrid.org

Scrödinger—Cycle Computing—USC

March 28, 2014 http://www.hpcwire.com

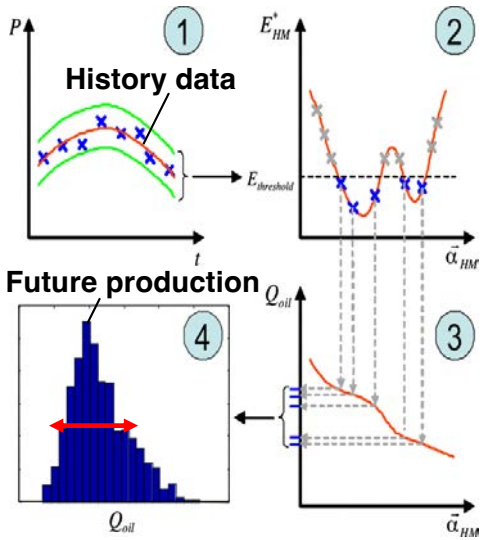
Schrödinger Partners with Cycle Computing to Accelerate Materials Simulation

NEW YORK, N.Y., March 28 — Schrödinger, LLC and Cycle Computing, LLC announced today a partnership that will allow customers to run Schrödinger's Materials Science Suite on the Cloud and elastic resources worldwide using Cycle Computing's CycleCloudä orchestration software. Cloud Computing provides users timely access to scalable computational resources as needed, without prohibitive upfront capital investment in infrastructure. Cycle Computing and Schrödinger have worked together on enabling many customer production workloads in the cloud, including the world's largest and fastest cloud computing run of more than 156,000 cores called the MegaRun in late 2013.

During the record-breaking MegaRun, Professor Mark Thompson at the University of Southern California (USC) completed the largest cloud-computing run in the world, using Schrödinger's software running on the CycleCloud platform. Professor Thompson calculated the optoelectronic properties for 205,000 materials with potential application in organic photovoltaic devices. The run used a maximum of 156,000+ CPU-cores completing 264 CPU-years of simulation in less than 18 hours.

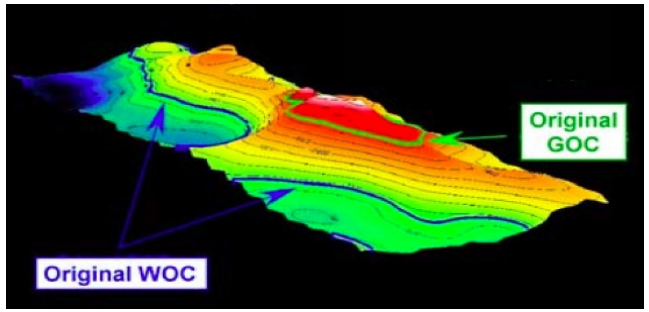
Parallel History Matching

- **Inverse problem: Calibration of reservoir simulation models to the observed production data**



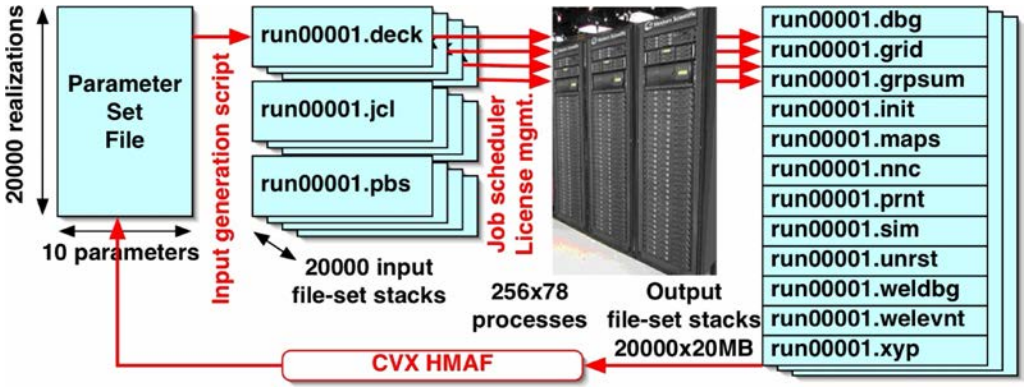
HMAF:
Landa & Guyaguler,
SPE 84465 ('03)

Parallel HMAF:
Landa, Kalia, Nakano, Nomura & Vashishta, *IPTC* 10751 ('05)

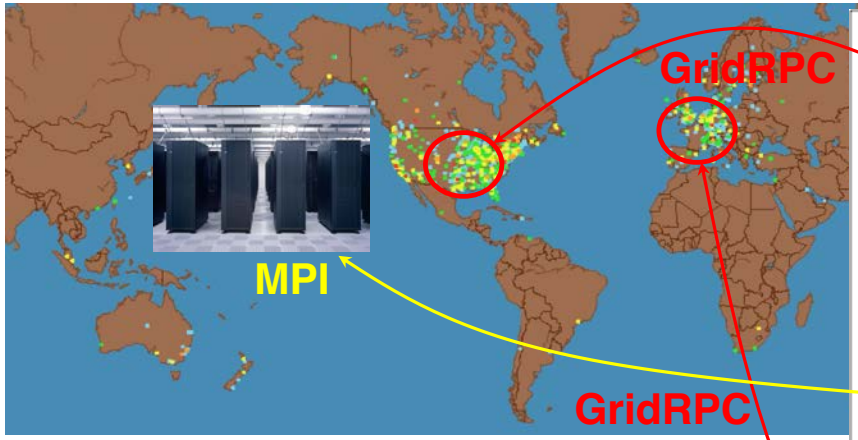


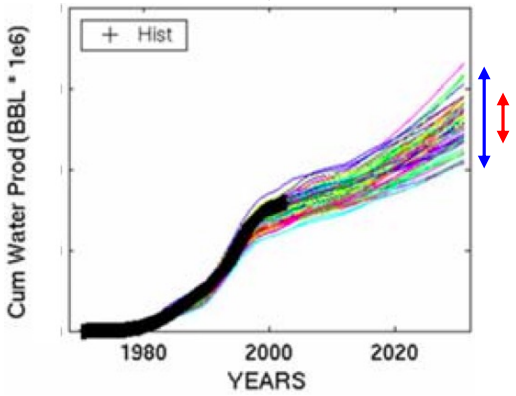
- Real field—offshore Africa
- 32 wells
- 30 years production history
- 30 years production forecast

- **CVX History Match & Associated Forecast (HMAF) framework: History matching & the assessment of uncertainties associated with flow prediction**

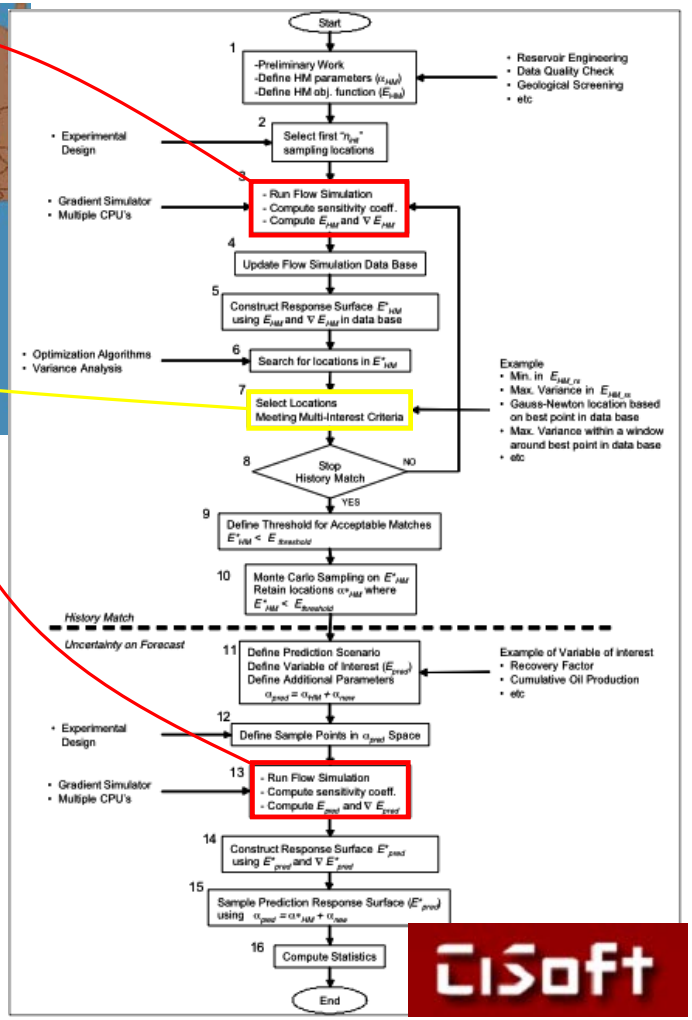


Overnight History Matching on a Grid





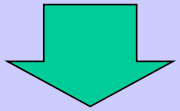
Nomura, Kalia, Nakano, Vashishta & Landa,
Journal of Supercomputing **41**, 109 ('07)



Grid Remote Procedure Call (RPC)

- **Simple RPC API (application program interface)**
- **Existing libraries & applications into Grid applications**
- **IDL (interface definition language) embodying call information, with minimal client-side management**

```
double A[n][n],B[n][n],C[n][n];      // Data Declaration
dmmul(n,A,B,C);                      // Call local function
```



```
grpc_function_handle_default(&hdl, "dmmul");
grpc_call(hdl,n,A,B,C);   // Call server side routine
```

- **Ninf–G Grid RPC system**
  http://ninf.apgrid.org

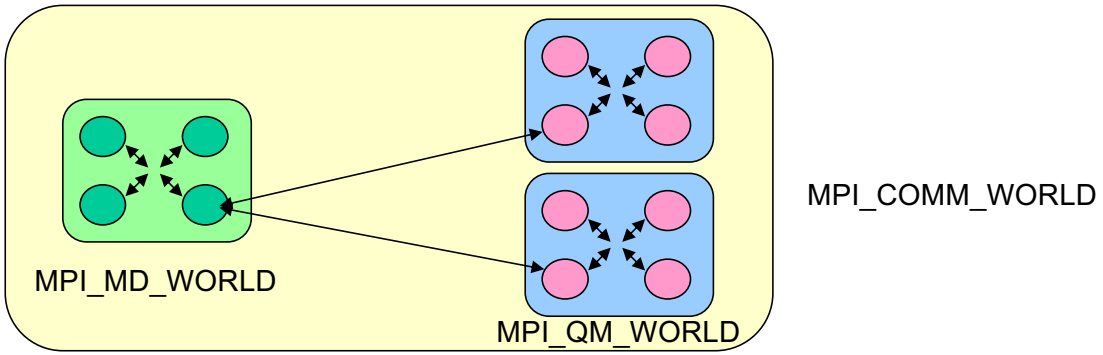


Outline

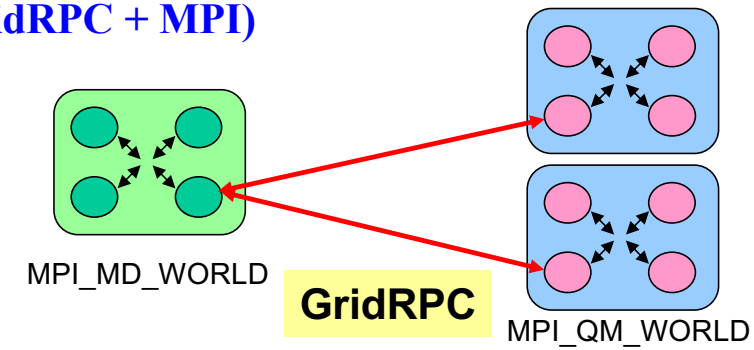
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Combined GridRPC+MPI MD/QM

- **Original implementation (MPICH-G2)**

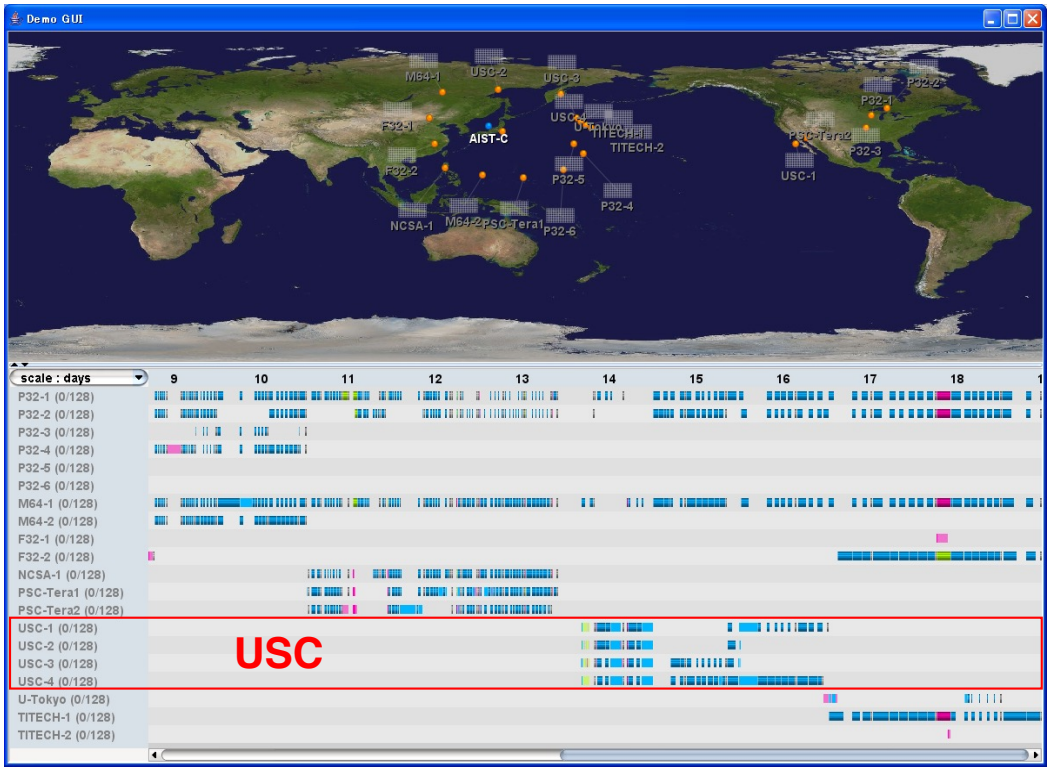


- **Hybrid implementation (GridRPC + MPI)**



- **Flexibility: Dynamically add/subtract, allocate, & migrate resources**
- **Fault tolerance: Automatically detect & recover from explicit (OS down or disconnected networks) & implicit (job stuck in a queue) faults**
- **Scalability: Manage 1000's of computing resources efficiently**

Global Grid QM/MD



H. Takemiya, Y. Tanaka, S. Sekiguchi (AIST)

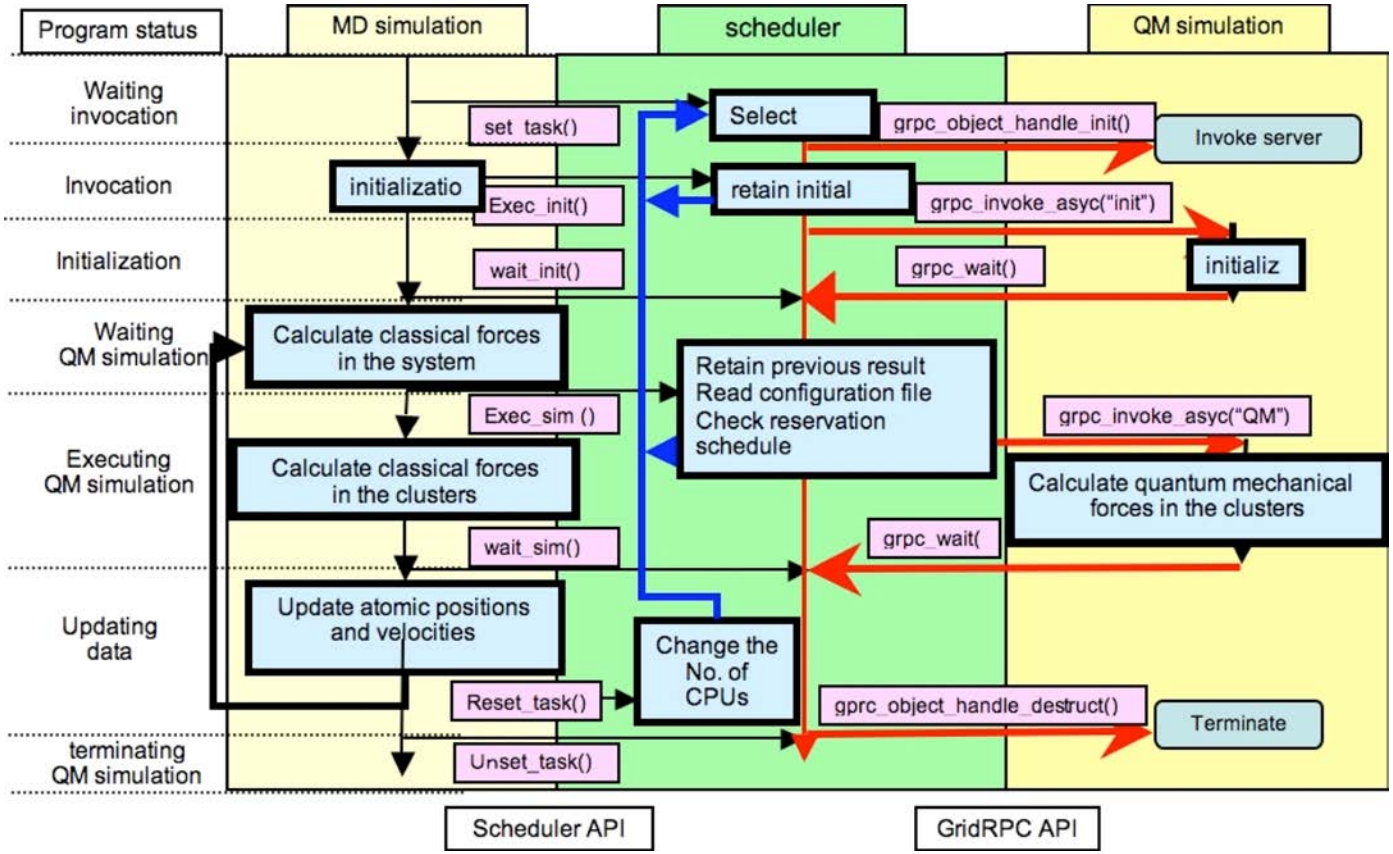
S. Ogata (NIT)

R.K. Kalia, A. Nakano, P. Vashishta (USC)

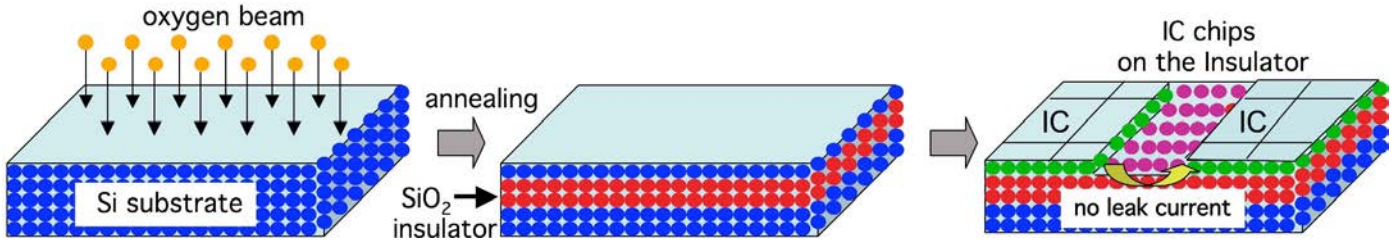
- **Hybrid GridRPC**(ninf.apgrid.org)**+MPI**(www.mcs.anl.gov/mpi) **Grid computing**
- **153,600 cpu-hrs metacomputing at 6 sites in the US (USC, PSC— Pittsburgh, NCSA— Illinois) & Japan (AIST, U Tokyo, TITech)**

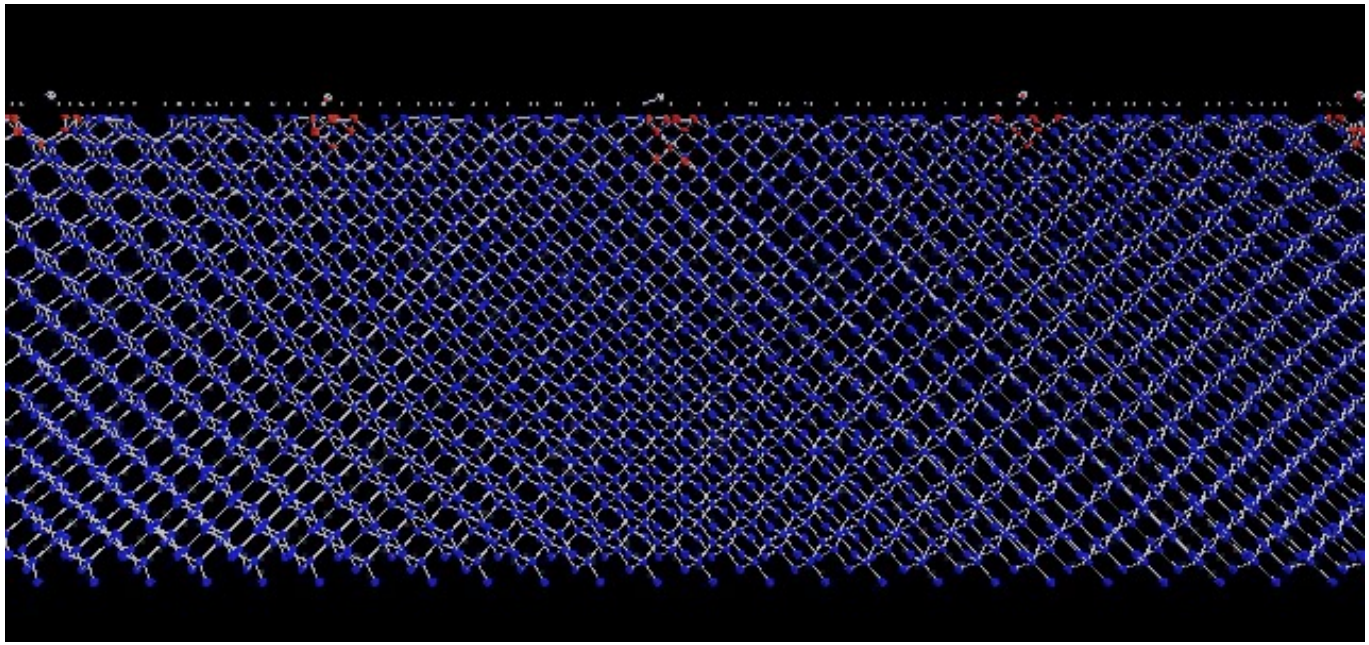
H. Takemiya *et al.*, *Proc. IEEE/ACM SC06*; Y. Song *et al.*, *Int'l J. Comput. Sci.* ('09); *CCGrid09*

Flow Chart of Grid MD/QM



SIMOX (Separation by Implantation by Oxygen)

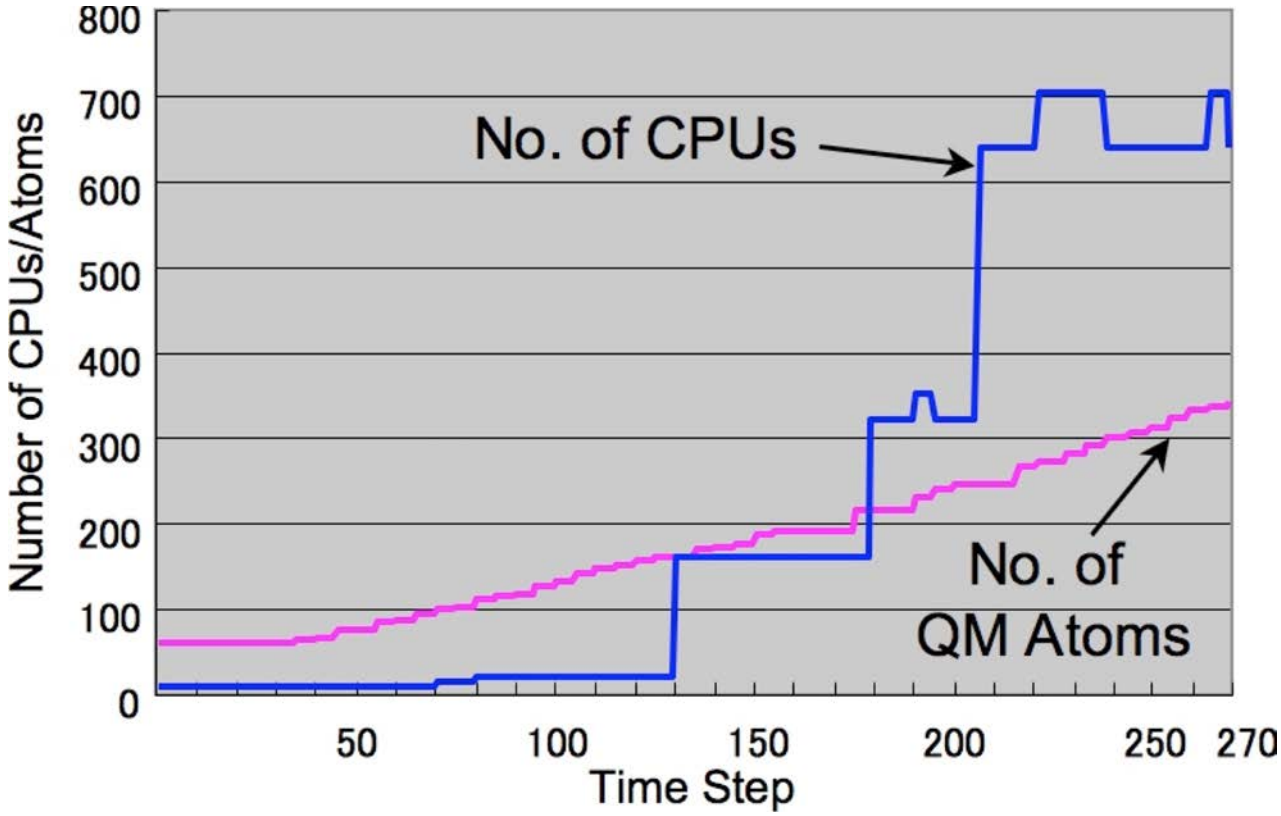




Red: quantum mechanically treated atoms ~ $O(N^3)$

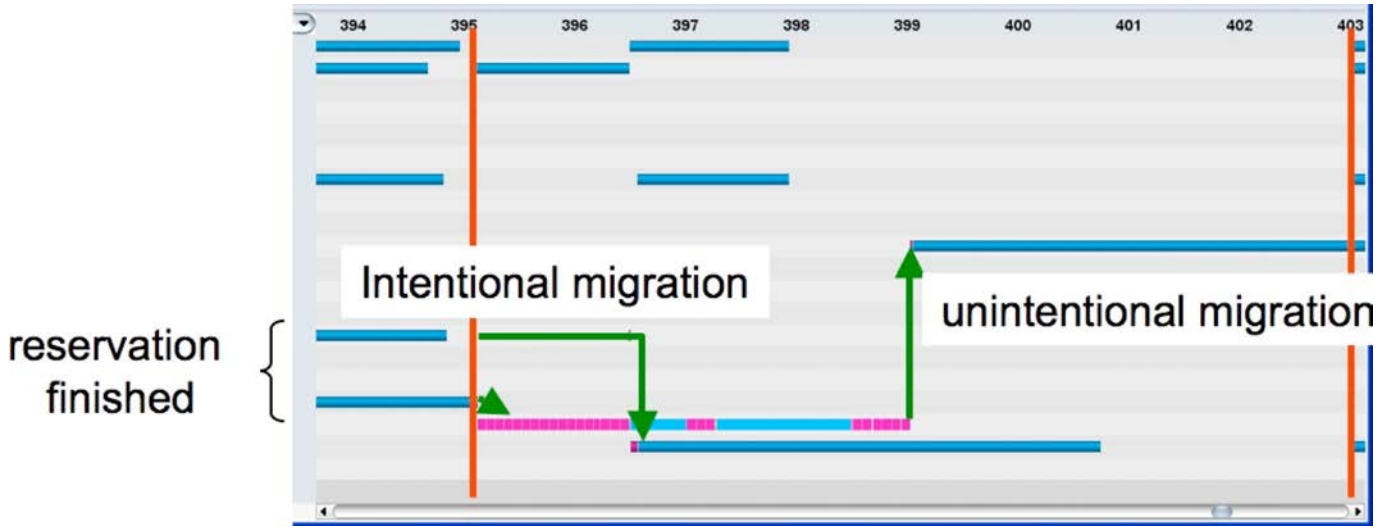
Flexibility: Adaptive MD/QM

- **Flexibility: Automated increase of the number of QM atoms on demand to maintain accuracy & associated dynamic re-allocation of CPUs**



Fault Tolerance

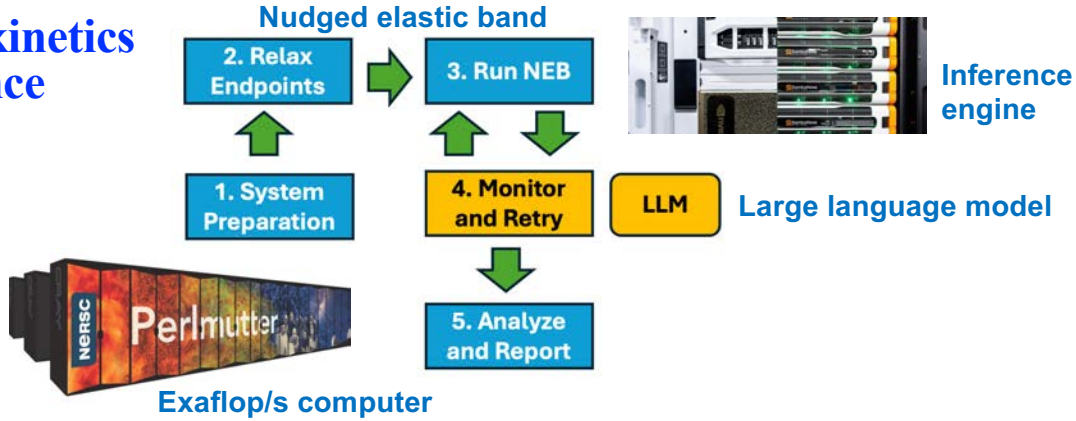
- **Automated migration in response to unexpected faults**



New: Exa- & Global-Scale Agentic AI Skill

- **Agentic skill = user's manual for AI agents: tacit (domain & procedural) knowledge**
- **Autonomous discovery of microkinetics on Exaflop/s computer & inference engine**

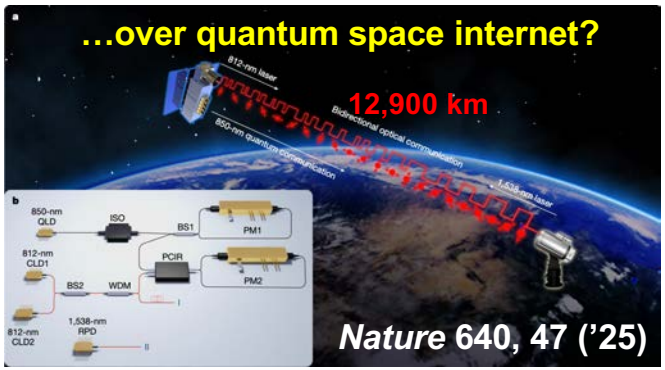
  Nomura *et al.*, *ACM AI Summit* ('26)
  doi: 10.48550/arXiv.2606.29100



- **Global metacomputing across Pacific**

  Takemiya *et al.*, *Supercomputing, SC06*, ieeexplore.ieee.org/document/4090197





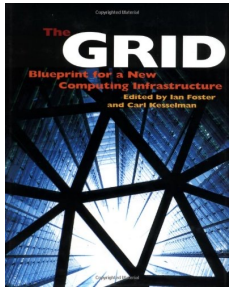
Outline

1. **Grid programming**
   - > **Metacomputing—multiscale MD/quantum-mechanical (QM) simulations: Grid-enabled MPI (MPI–G2)**
   - > **Task farm: Grid remote procedure call (Ninf–G)**
   - > **Sustainable & adaptive Grid supercomputing**

2. **Grid software (1998-2018)***
   - > **Globus toolkit**
   - > **Open Grid Services Architecture (OGSA)**

Ian Foster (ANL)







Carl Kesselman (USC)

*USC-CARC uses Globus for file transfer

https://www.carc.usc.edu/user-guides/research-data-management/transferring-data/transferring-data-globus

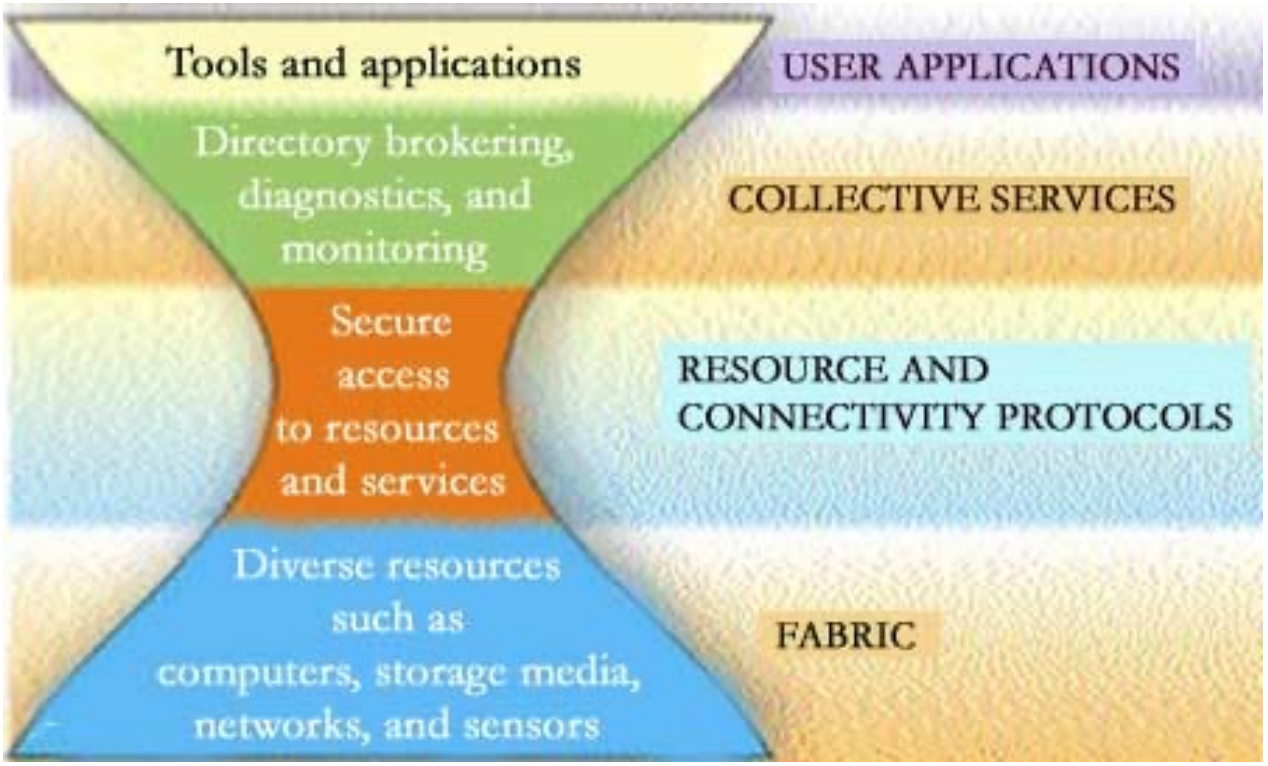
Grid System

Goal: Coordinated resource sharing & problem solving in dynamic, multi-institutional virtual organizations

1. **Coordinate distributed resources**
2. **Use standard, open, general-purpose protocols & interfaces**
3. **Deliver nontrivial qualities of services (QoS)**

Grid Architecture

Layered architecture



- **Hourglass model: A small set of core protocols (*e.g.*, TCP/IP) + various: (1) high-level behaviors & (2) underlying technologies**

Layered Grid Architecture

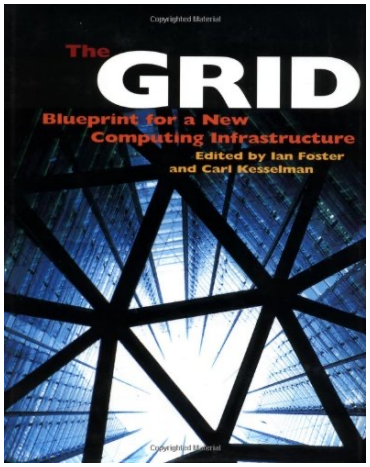
- **Fabric:** Introspection & management of local resources
  - > **Computational resources:** Start programs & monitor/control the execution of the resulting processes
  - > **Storage:** Put & get files (*e.g.*, disk space allocation)
  - > **Network:** Control network transfers (*e.g.*, prioritization)
- **Connectivity:** Define communication & authentication protocols
- **Resource:** Define protocols for negotiation, initiation, monitoring, control, accounting & payment of sharing operations on individual resources
- **Collective:** Capture interactions across collection of resources, *e.g.*, directory services, co-allocation & data replication

Globus Toolkit (1998-2018)

- **Globus Toolkit version 2 (GT2): Open source, de facto standard of Grid computing middleware to construct interoperable Grid applications ('97)**
  - > **Define & implement protocols, application program interfaces (APIs) & services**
  - > **Provide solutions to authentication, resource discovery & resource access**
- **Globus Toolkit version 6 (GT6) at**
  http://www.globus.org

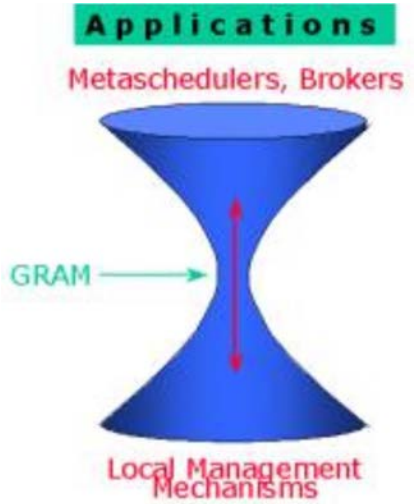


globus



GT2: Globus Toolkit 2

- **Fabric**
  - > **General purpose architecture for reservation & allocation (GARA)**
- **Connectivity**
  - > **Grid security infrastructure (GSI)**
- **Resource**
  - > **Grid resource allocation & management (GRAM) protocol**



Outline

1. Grid programming
   - > Metacomputing—multiscale MD/quantum-mechanical (QM) simulations: Grid-enabled MPI (MPI–G2)
   - > Task farm: Grid remote procedure call (Ninf–G)
   - > Sustainable & adaptive Grid supercomputing
2. Grid software
   - > Globus toolkit
   - > Open Grid Services Architecture (OGSA)

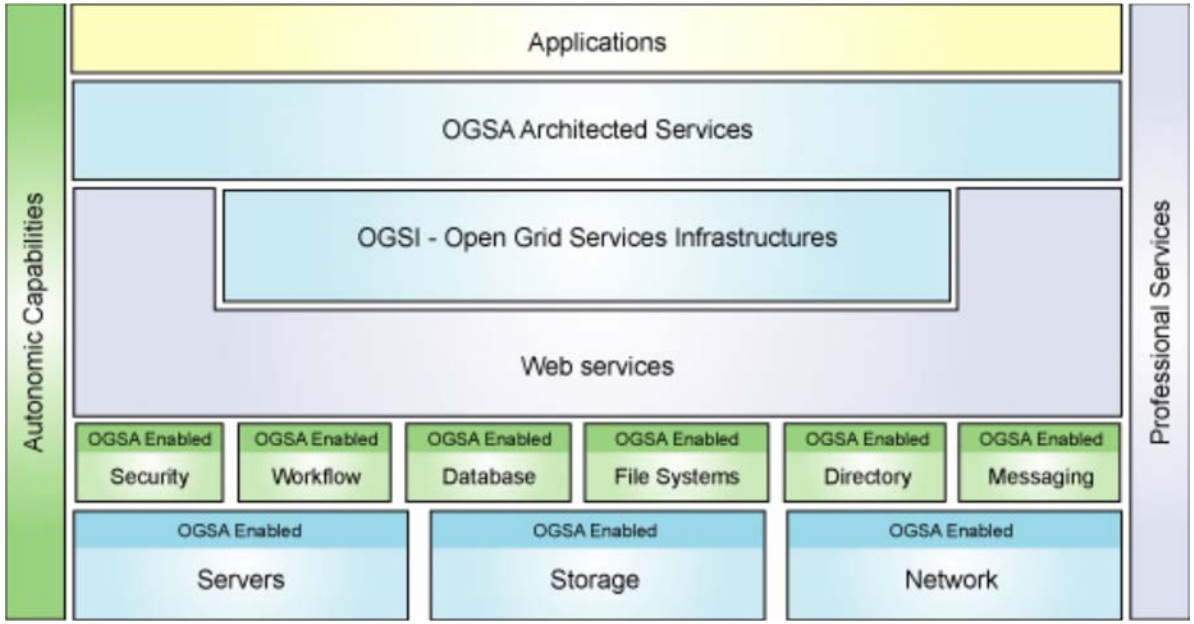
Open Grid Services Architecture (OGSA)

- **OGSA = Definition of a service-oriented infrastructure**
- **Service: A network-enabled entity with a well-defined interface that provides some capability**

1. **Align Grid computing with industrial initiatives in service-oriented architecture & Web services**
2. **Provide a framework within which to define interoperable & portable services**
3. **Define a core set of standard interfaces & behaviors**
4. **Implemented in the OGSA-based Globus Toolkit 3 (currently GT 6.0):**
   http://www.globus.org

Open Grid Services Architecture (OGSA)

1. **Web services description language (WSDL): An interface definition language describing services (or software components) independent of platforms**
2. **Open grid services infrastructure (OGSI): A set of WSDL interfaces & associated conventions, extensions & refinements to Web services standards to support basic Grid**



OGSI Functionalities

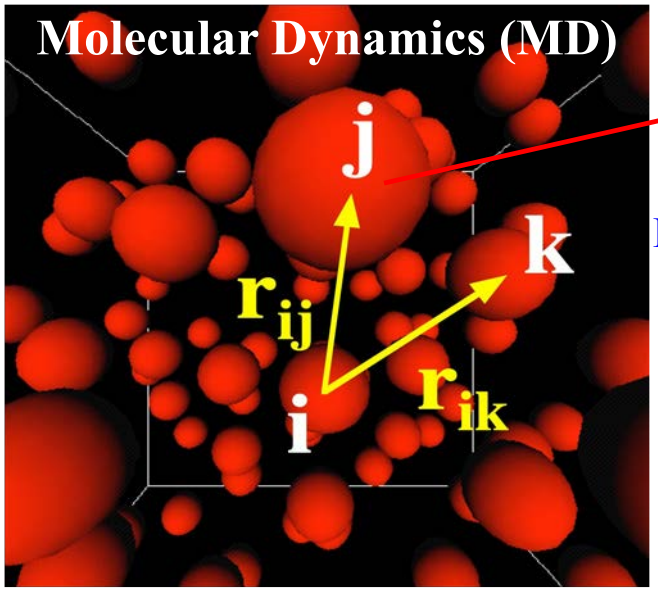
- **Grid service description & instances: Definition & execution**
- **Service state, metadata & introspection**
- **Naming & name resolution: Universal resource identifier (URI)**
- **Service life cycle: Instantiate & destruct**
- **Fault type: Standard base type for all fault messages**
- **Service groups: Represent & manage groups of services**

OGSA Services

- **Core services**
  - > **Name resolution & discovery**
  - > **Security**
  - > **Policy**
  - > **Messaging, queuing & logging**
  - > **Events**
  - > **Metering & accounting: Resource usage & charges**
- **Data & information services**
  - > **Data management & access**
  - > **Replication**
  - > **Metadata & provenance**
- **Resource & service management**
  - > **Provisioning & resource management**
  - > **Service orchestration**

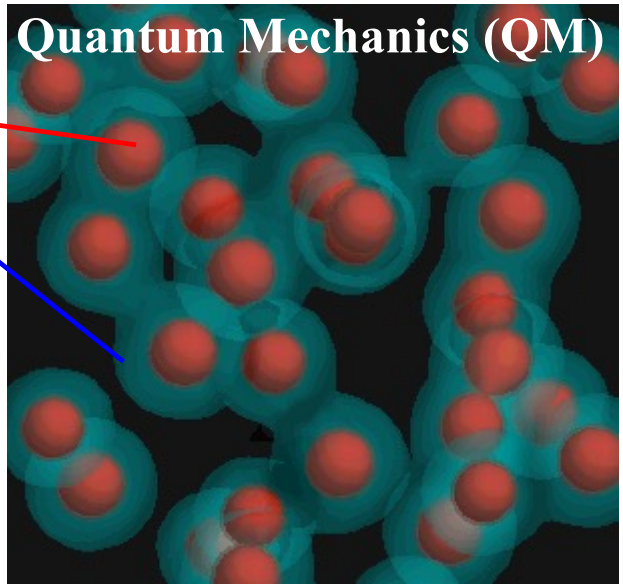
Virtualization-aware Application Framework

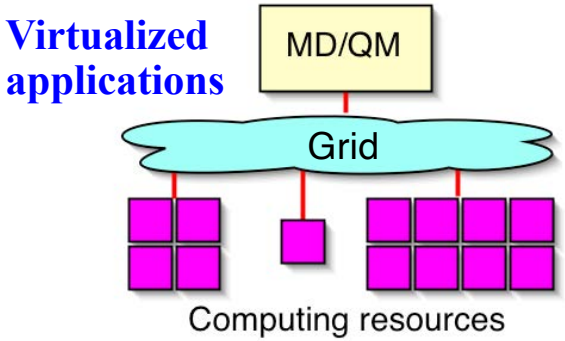
Atomistic materials simulation methods



Atom

Electron density





- Scalability
- Portable performance
- Adaptation

→ Data-locality principles
Divide-conquer-recombine

doi: 10.1142/9789812701084_0005

Research Issues

1. **Computational complexity:** Computation time, T, as a function of the problem size, N

   $O(N) < O(N^m) < O(C^N)$

   Easy ← Hard

2. **Scalability:** Parallel efficiency

   $\eta = T_1/(T_p p) \sim 1$

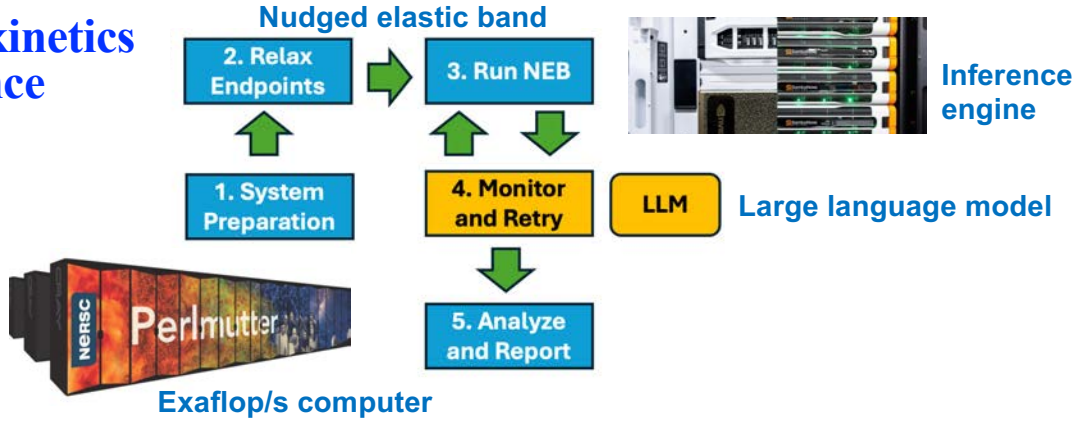
   for a large number, p, of processors

3. **Fault resilience**

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